

FarmLink — Requirements & Field Reference

Single source of truth. Backend and Frontend MUST match these names exactly.

Stack

Layer	Technology
Backend	Java 17 + Spring Boot 3.x
Database	PostgreSQL 16 + PostGIS 3.4
Storage	MinIO (S3-compatible)
Frontend	React 18 + Vite
Auth	JWT (Bearer token, 24h expiry)
Container	Docker Compose (DB + MinIO only)

Environment Variables (canonical list)

See `.env` file. Spring Boot reads via `application.yml`.
Frontend reads via `VITE_` prefix equivalents.

Auth

- Login: `mobile` + `password`
- Registration collects: `full_name`, `mobile`, `password`, `aadhaar_number`, `address_line`, `city`, `state`, `pincode`, `latitude`, `longitude`
- Token: JWT Bearer, header `Authorization: Bearer <token>`

DB Tables & Key Fields

users

Field	Type	Notes
id	bigint PK	
full_name	varchar(120)	
mobile	varchar(15)	unique, login ID
password_hash	varchar(255)	bcrypt
aadhaar_number	varchar(12)	unique
address_line	varchar(300)	
city	varchar(100)	
state	varchar(100)	
pincode	varchar(10)	
location	geography(point,4326)	PostGIS
latitude	double	denormalised
longitude	double	denormalised
profile_picture_url	varchar(500)	
is_active	boolean	default true

equipment_listings

Field	Type	Notes
id	bigint PK	
owner_id	bigint FK → users	
equipment_type	varchar(100)	"Plough","Seeder", etc.
description	text	optional
rental_basis	enum	HOURLY DAILY BOTH

Field	Type	Notes
hourly_rate	numeric(10,2)	null if DAILY only
daily_rate	numeric(10,2)	null if HOURLY only
security_deposit	numeric(10,2)	default 0
available_from	date	
available_till	date	
status	enum	ACTIVE BOOKED EXPIRED DELETED
location	geography	copied from owner at create time
latitude	double	
longitude	double	

equipment_images

Field	Type	Notes
id	bigint PK	
listing_id	bigint FK → equipment_listings	
image_url	varchar(500)	full URL for display
object_key	varchar(300)	MinIO key for deletion
display_order	smallint	0-based

equipment_rental_requests

Field	Type	Notes
id	bigint PK	
listing_id	bigint FK → equipment_listings	
requester_id	bigint FK → users	
rental_basis	enum	HOURLY or DAILY

Field	Type	Notes
from_date	date	DAILY only
to_date	date	DAILY only
num_hours	smallint	HOURLY only
description	text	requester's note
status	enum	PENDING ACCEPTED REJECTED CANCELLED
requested_at	timestamptz	
responded_at	timestamptz	

Business rules:

- When owner accepts one request → all other PENDING requests for same listing → auto REJECTED
- Listing status → BOOKED
- Once BOOKED → listing hidden from Rent Equipment page

operator_profiles

Field	Type	Notes
id	bigint PK	
user_id	bigint FK → users	unique (1 profile per user)
services_offered	text[]	["SEEDING", "PLOUGHING", ...]
profile_picture_url	varchar(500)	
profile_picture_key	varchar(300)	MinIO key
hourly_rate	numeric(10,2)	
daily_rate	numeric(10,2)	
is_rate_negotiable	boolean	
bio	text	
status	enum	ACTIVE BOOKED EXPIRED DELETED

Field	Type	Notes
location	geography	
latitude	double	
longitude	double	

Allowed service values: SEEDING, PLOUGHING, HARVESTING, TRANSPORT, LEVELING, OTHER

operator_hire_requests

Field	Type	Notes
id	bigint PK	
operator_id	bigint FK → operator_profiles	
requester_id	bigint FK → users	
from_date	date	
to_date	date	
offered_rate	numeric(10,2)	what requester willing to pay
rate_basis	enum	HOURLY or DAILY
description	text	
status	enum	PENDING ACCEPTED REJECTED CANCELLED
requested_at	timestampz	
responded_at	timestampz	

Business rules:

- Accept one → auto reject others for same operator same dates (app-layer check)
- Operator profile status → BOOKED until to_date passes

job_posts

Field	Type	Notes
id	bigint PK	
posted_by	bigint FK → users	
title	varchar(200)	
description	text	
service_needed	varchar(100)	SEEDING, PLOUGHING, etc.
desired_rate	numeric(10,2)	farmer's budget
rate_basis	enum	HOURLY or DAILY
work_from_date	date	
work_to_date	date	
status	enum	ACTIVE BOOKED EXPIRED DELETED
location	geography	
latitude	double	
longitude	double	

job_applications

Field	Type	Notes
id	bigint PK	
job_post_id	bigint FK → job_posts	
applicant_id	bigint FK → users	
offered_rate	numeric(10,2)	operator's asking rate
rate_basis	enum	HOURLY or DAILY
cover_note	text	
status	enum	PENDING ACCEPTED REJECTED CANCELLED

Field	Type	Notes
applied_at	timestampz	
responded_at	timestampz	
UNIQUE	(job_post_id, applicant_id)	one application per operator per job

notifications

Field	Type	Notes
id	bigint PK	
recipient_id	bigint FK → users	
type	enum	see notification_type enum
message	text	human-readable
reference_id	bigint	ID in relevant table
is_read	boolean	default false
created_at	timestampz	

Notification types: `RENTAL_REQUEST_RECEIVED`, `RENTAL_REQUEST_ACCEPTED`, `RENTAL_REQUEST_REJECTED`, `HIRE_REQUEST_RECEIVED`, `HIRE_REQUEST_ACCEPTED`, `HIRE_REQUEST_REJECTED`, `JOB_APPLICATION_RECEIVED`, `JOB_APPLICATION_ACCEPTED`, `JOB_APPLICATION_REJECTED`

MinIO Buckets

Bucket	Content
<code>equipment-images</code>	Equipment listing photos
<code>operator-images</code>	Operator profile pictures

Object key format: `{bucket}/{userId}/{uuid}.{ext}`

- Default radius: **10 km** (configurable via `DEFAULT_SEARCH_RADIUS_KM` env)
- User can expand up to `MAX_SEARCH_RADIUS_KM` (100 km)
- PostGIS query pattern:

```
sql

WHERE ST_DWithin(location, ST_MakePoint(:lng, :lat)::geography, :radiusMeters)
```

API Endpoint Map (to be implemented)

Auth

- `POST /api/auth/register`
- `POST /api/auth/login`

Equipment

- `GET /api/equipment` — list (geo-filtered, paginated)
- `POST /api/equipment` — create listing (+ image upload)
- `GET /api/equipment/{id}` — single listing
- `DELETE /api/equipment/{id}` — owner deletes
- `POST /api/equipment/{id}/request` — send rental request
- `GET /api/equipment/requests/received` — owner sees incoming requests
- `GET /api/equipment/requests/sent` — user sees sent requests
- `PATCH /api/equipment/requests/{id}/accept`
- `PATCH /api/equipment/requests/{id}/reject`

Operators

- `POST /api/operators` — create profile
- `GET /api/operators` — list (geo-filtered)
- `GET /api/operators/{id}` — single profile
- `POST /api/operators/{id}/hire` — send hire request
- `GET /api/operators/requests/received` — operator sees incoming
- `GET /api/operators/requests/sent` — user sees sent
- `PATCH /api/operators/requests/{id}/accept`

- PATCH /api/operators/requests/{id}/reject

Jobs

- GET /api/jobs — list (geo-filtered)
- POST /api/jobs — post a job
- GET /api/jobs/{id}
- POST /api/jobs/{id}/apply — operator applies
- GET /api/jobs/applications/received — farmer sees applicants
- GET /api/jobs/applications/sent — operator sees sent
- PATCH /api/jobs/applications/{id}/accept
- PATCH /api/jobs/applications/{id}/reject

Notifications

- GET /api/notifications — all for logged-in user
- PATCH /api/notifications/{id}/read
- PATCH /api/notifications/read-all